

CHEMISTRY 30

UNIT B — ELECTROCHEMICAL CHANGES

ALBERTA DIPLOMA PREPARATION STUDY GUIDE

Mechanism • Calculations • Validation • Diploma-style writing

 Theme: Change & Energy — Electrochemical Change

 Focus: Redox + cells + E° + electrolysis + Faraday + titration

 Includes: objectives, checklist, 10 fully worked diploma questions

1 UNIT / MODULE OBJECTIVES

Aligned to Alberta Chemistry 30 Unit B outcomes (30–B1 and 30–B2).

Unit Objectives (measurable diploma verbs)

- Analyze oxidation and reduction both operationally (electron transfer) and theoretically (oxidation number change). (30–B1.1k)
- Define and correctly use: oxidizing agent, reducing agent, oxidation number, half-reaction, disproportionation; and electrochemical cell vocabulary (anode, cathode, electrolyte, salt bridge/porous cup, external circuit, power supply, voltaic cell, electrolytic cell). (30–B1.2k; 30–B2.1k)
- Differentiate redox reactions from other reactions using oxidation numbers and/or half-reactions. (30–B1.3k)
- Identify electron transfer and agents in real contexts (cellular respiration/photosynthesis; corrosion). (30–B1.4k)
- Compare strengths of oxidizing/reducing agents using empirical data and standard reduction potentials. (30–B1.5k)
- Predict spontaneity using standard reduction potentials and compare predictions to results; explain discrepancies. (30–B1.6k; 30–B2.3s)

- Write and balance redox equations in acidic and neutral solutions using half-reactions (from SRP table or derived). (30–B1.7k)
- Perform calculations for redox titrations (quantitative redox stoichiometry). (30–B1.8k)
- Predict and write half-reactions at each electrode; recognize predicted reactions do not always occur (e.g., brine electrolysis). (30–B2.3k; 30–B2.4k)
- Explain SRP values are relative to the hydrogen electrode (0 V) at standard conditions; calculate E°_{cell} and use E°_{cell} to predict spontaneity. (30–B2.5k–30–B2.7k)
- Apply Faraday's law + stoichiometry to calculate mass/amount/current/time in voltaic and electrolytic cells. (30–B2.8k)
- Evaluate STS applications: batteries, electroplating, refining/electrowinning, pool chlorination, corrosion control, and fuel cells. (30–B2.1sts–30–B2.3sts)

Concept / Skill	Diploma-Level Expectation
Oxidation/Reduction	Use oxidation numbers AND half-reactions; always name oxidizing and reducing agents with evidence.
Spontaneity	Use E°_{cell} sign and SRP positions; justify with electron-flow logic.
Balancing Redox	Acidic/neutral: atoms $\rightarrow$ O with H_2O $\rightarrow$ H with H^+/OH^- $\rightarrow$ charge with e^- $\rightarrow$ equalize e^- $\rightarrow$ combine.
Electrolysis	Predict products carefully; in aqueous solutions, water can compete; justify using SRP + kinetics/context.
Faraday's Law	$Q = It$; $n(\text{e}^-) = Q/F$; use stoichiometry to link e^- to products; track units & sig figs.

2 STUDENT SELF-VERIFICATION CHECKLIST

Check each statement only if you can do it independently and explain your reasoning (diploma standard).

Self-Verification Checklist

Vocabulary precision

- ☐ I can define oxidation and reduction operationally (electron transfer) and theoretically (oxidation number change) and give an example of each.
- ☐ I can define oxidizing agent and reducing agent and justify which reactant is which by referencing electron loss/gain.
- ☐ I can define anode/cathode and apply 'oxidation at anode, reduction at cathode' in BOTH voltaic and electrolytic cells.
- ☐ I can define standard reduction potential and explain why the hydrogen electrode is 0 V by convention.

Mechanistic understanding

- ☐ I can trace electron flow, ion migration, and charge balance in a voltaic cell diagram and explain why a salt bridge is required.
- ☐ I can compare how energy is transformed in voltaic (chemical $\rightarrow$ electrical) vs electrolytic (electrical $\rightarrow$ chemical) cells.
- ☐ I can predict competing reactions in aqueous electrolysis and justify the actual products using SRP and context (e.g., brine).

Quantitative reasoning

- ☐ I can compute E°_{cell} correctly from SRP data and interpret its sign to predict spontaneity.
- ☐ I can solve Faraday problems involving multiple steps ($I \leftrightarrow t \leftrightarrow Q \leftrightarrow \text{mol } e^- \leftrightarrow \text{mol product} \leftrightarrow \text{mass}$) with correct units.
- ☐ I can complete a redox titration calculation by extracting the mole ratio from a balanced redox equation.

Data & error analysis

- ☐ I can compare predicted vs observed cell potential and propose at least two realistic reasons for discrepancies (nonstandard concentration, overpotential, internal resistance).
- ☐ I can recognize common misconceptions (e.g., 'anode is always negative' / 'electrons travel through salt bridge') and correct them.

STS / evaluation

- ☐ I can evaluate a corrosion-prevention strategy (galvanizing, sacrificial anode, impressed current) using redox reasoning and SRP evidence.
- ☐ I can make an evidence-based recommendation about battery/fuel cell use considering efficiency, cost, and environmental impacts.

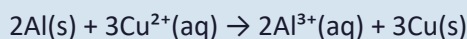
3 DIPLOMA-LEVEL PRACTICE QUESTIONS WITH FULLY WORKED SOLUTIONS

Exactly 10 questions. Every solution includes: Step 1 Identification → Step 2 Principle → Step 3 Explicit Reasoning → Step 4 Validation → Step 5 Final Answer.

Question 1

Student Task

Assign oxidation numbers and identify the oxidizing agent and reducing agent for:



Then state which species is oxidized and which is reduced.

Fully Worked Solution

Step 1 — Identification: Al(s) is element → ON = 0. Cu²⁺ has ON = +2. Products: Al³⁺ has ON = +3; Cu(s) has ON = 0.

Step 2 — Principle Statement: Oxidation = increase in oxidation number (loss of e⁻). Reduction = decrease in oxidation number (gain of e⁻). Oxidizing agent is reduced; reducing agent is oxidized.

Step 3 — Explicit Reasoning: Al: 0 → +3 (increase) ⇒ Al is oxidized (Al loses e⁻). Cu: +2 → 0 (decrease) ⇒ Cu²⁺ is reduced (Cu²⁺ gains e⁻).

Oxidizing agent: Cu²⁺ (because it is reduced). Reducing agent: Al(s) (because it is oxidized).

Step 4 — Validation: Charges balance: 3 Cu²⁺ implies 6+ charge; 2 Al³⁺ implies 6+ charge; consistent electron transfer.

Step 5 — Final Answer: Al is oxidized; Cu²⁺ is reduced; oxidizing agent = Cu²⁺; reducing agent = Al(s).

Common Error

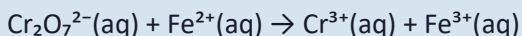
Calling the oxidizing agent the species that 'oxidizes' others (word trap) rather than the species that is reduced.

Exam Strategy

Write two statements every time: 'oxidizing agent is reduced' and 'reducing agent is oxidized,' then match to oxidation-number changes.

Question 2**Student Task**

Balance the following redox equation in acidic solution using the half-reaction method:



Show all steps and electron balancing.

Fully Worked Solution

Step 1 — Identification: Cr goes from +6 in $\text{Cr}_2\text{O}_7^{2-}$ to +3 in Cr^{3+} (reduction). Fe^{2+} to Fe^{3+} is oxidation.

Step 2 — Principle Statement: Half-reaction method (acidic): balance atoms (except H,O) $\rightarrow$ balance O with H_2O $\rightarrow$ balance H with H^+ $\rightarrow$ balance charge with e^- $\rightarrow$ equalize e^- $\rightarrow$ add and cancel.

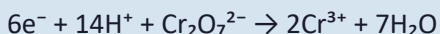
Step 3 — Explicit Reasoning:

Reduction half-reaction: $\text{Cr}_2\text{O}_7^{2-} \rightarrow 2\text{Cr}^{3+}$

Balance O by adding $7\text{H}_2\text{O}$ to products: $\text{Cr}_2\text{O}_7^{2-} \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$

Balance H by adding 14H^+ to reactants: $14\text{H}^+ + \text{Cr}_2\text{O}_7^{2-} \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$

Balance charge with e^- : Left charge = $+14 - 2 = +12$. Right charge = $+6$. Add 6e^- to left to reduce charge:



Oxidation half-reaction: $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$

Equalize electrons: multiply oxidation by 6: $6\text{Fe}^{2+} \rightarrow 6\text{Fe}^{3+} + 6\text{e}^-$

Add and cancel 6e^- : $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{Fe}^{2+} \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O} + 6\text{Fe}^{3+}$

Step 4 — Validation: Atom check: Cr 2=2; O 7=7; H 14=14; Fe 6=6. Charge: left $(-2 + 14 + 12) = +24$? careful: $14\text{H}^+ = +14$, $6\text{Fe}^{2+} = +12$, plus $-2 = +24$. Right: $2\text{Cr}^{3+} = +6$, $6\text{Fe}^{3+} = +18$ total $+24$. Balanced.

Step 5 — Final Answer: $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{Fe}^{2+} \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O} + 6\text{Fe}^{3+}$.

Common Error

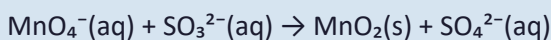
Balancing atoms but forgetting to balance overall charge with electrons (leading to an incorrect stoichiometric ratio).

Exam Strategy

After balancing atoms, always do a charge audit on each half-reaction and fix with e^- before combining.

Question 3**Student Task**

Balance the following redox equation in neutral/basic solution (use OH^- method):



Show half-reactions and the conversion from acidic to basic/neutral conditions.

Fully Worked Solution

Step 1 — Identification: Mn goes +7 in MnO_4^- to +4 in MnO_2 (reduction). S goes +4 in SO_3^{2-} to +6 in SO_4^{2-} (oxidation).

Step 2 — Principle Statement: Balance as if acidic (H_2O/H^+), then convert H^+ to OH^- by adding OH^- to both sides ($H^+ + OH^- \rightarrow H_2O$) and simplify.

Step 3 — Explicit Reasoning:

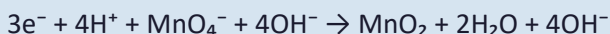
Reduction (acidic first): $MnO_4^- \rightarrow MnO_2$

Balance O: left has 4 O, right has 2 O $\rightarrow$ add $2H_2O$ to right? Actually add $2H_2O$ to products gives $MnO_4^- \rightarrow MnO_2 + 2H_2O$

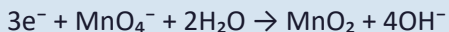
Balance H with H^+ : add $4H^+$ to reactants: $4H^+ + MnO_4^- \rightarrow MnO_2 + 2H_2O$

Balance charge with e^- : left charge = $+4 - 1 = +3$; right = 0. Add $3e^-$ to left: $3e^- + 4H^+ + MnO_4^- \rightarrow MnO_2 + 2H_2O$

Convert to basic: add $4OH^-$ to both sides to eliminate $4H^+$:



Combine $4H^+ + 4OH^- \rightarrow 4H_2O$, then cancel waters: left $4H_2O$, right $2H_2O \rightarrow$



Oxidation (acidic first): $SO_3^{2-} \rightarrow SO_4^{2-}$

Balance O: add H_2O to reactants: $H_2O + SO_3^{2-} \rightarrow SO_4^{2-}$

Balance H: add $2H^+$ to products: $H_2O + SO_3^{2-} \rightarrow SO_4^{2-} + 2H^+$

Balance charge with e^- : left charge = -2 ; right = $-2 + 2 = 0 \rightarrow$ add $2e^-$ to products to make right -2 :



Convert to basic: add 2OH^- to both sides: $\text{H}_2\text{O} + \text{SO}_3^{2-} + 2\text{OH}^- \rightarrow \text{SO}_4^{2-} + 2\text{H}^+ + 2\text{e}^- + 2\text{OH}^-$

$2\text{H}^+ + 2\text{OH}^- \rightarrow 2\text{H}_2\text{O}$; simplify: $\text{H}_2\text{O} + \text{SO}_3^{2-} + 2\text{OH}^- \rightarrow \text{SO}_4^{2-} + 2\text{H}_2\text{O} + 2\text{e}^-$

Cancel 1 H_2O : $\text{SO}_3^{2-} + 2\text{OH}^- \rightarrow \text{SO}_4^{2-} + \text{H}_2\text{O} + 2\text{e}^-$

Equalize electrons: LCM of 3 and 2 is 6. Multiply Mn half by 2 and S half by 3:

Mn half $\times 2$: $6\text{e}^- + 2\text{MnO}_4^- + 4\text{H}_2\text{O} \rightarrow 2\text{MnO}_2 + 8\text{OH}^-$

S half $\times 3$: $3\text{SO}_3^{2-} + 6\text{OH}^- \rightarrow 3\text{SO}_4^{2-} + 3\text{H}_2\text{O} + 6\text{e}^-$

Add and cancel 6e^- ; simplify OH^- and H_2O :

$2\text{MnO}_4^- + 4\text{H}_2\text{O} + 3\text{SO}_3^{2-} + 6\text{OH}^- \rightarrow 2\text{MnO}_2 + 8\text{OH}^- + 3\text{SO}_4^{2-} + 3\text{H}_2\text{O}$

Cancel 3 H_2O (right) with 4 H_2O (left) leaves 1 H_2O on left; cancel 6 OH^- (left) with 8 OH^- (right) leaves 2 OH^- on right:

$2\text{MnO}_4^- + \text{H}_2\text{O} + 3\text{SO}_3^{2-} \rightarrow 2\text{MnO}_2 + 3\text{SO}_4^{2-} + 2\text{OH}^-$

Step 4 — Validation: Atom/charge check: O and H conserved; net charge left = $(2 \times -1) + (3 \times -2) = -2 - 6 = -8$.

Right = $(3 \times -2) + (2 \times -1) = -6 - 2 = -8$. Balanced.

Step 5 — Final Answer: $2\text{MnO}_4^- + \text{H}_2\text{O} + 3\text{SO}_3^{2-} \rightarrow 2\text{MnO}_2 + 3\text{SO}_4^{2-} + 2\text{OH}^-$.

Common Error

Leaving H^+ in the final answer for a basic/neutral question (conversion step missing).

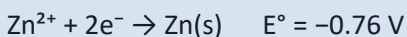
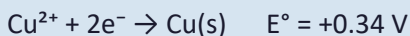
Exam Strategy

After balancing in acidic form, *always* eliminate H^+ by adding OH^- and simplifying waters before finalizing.

Question 4

Student Task

Use the following standard reduction potentials (E°):



(a) Identify anode/cathode in a spontaneous cell built from these half-cells.

(b) Calculate E°_{cell} .

(c) Write the overall balanced redox equation.

Fully Worked Solution

Step 1 — Identification: Higher E° reduction occurs as reduction at cathode. Cu^{2+}/Cu has higher E° (+0.34) than Zn^{2+}/Zn (−0.76).

Step 2 — Principle Statement: In a spontaneous voltaic cell, cathode is the half-reaction with higher E°_{red} .
 $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$ (using reduction potentials).

Step 3 — Explicit Reasoning: Cathode (reduction): $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$. Anode is oxidation of Zn: $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ (reverse of listed reduction).

$$E^\circ_{\text{cell}} = 0.34 - (-0.76) = +1.10 \text{ V}.$$

Overall equation (electrons cancel as written): $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$.

Step 4 — Validation: E°_{cell} positive $\Rightarrow$ spontaneous (consistent). Charge and atoms conserved.

Step 5 — Final Answer: Cathode = Cu^{2+}/Cu , Anode = Zn/Zn^{2+} ; $E^\circ_{\text{cell}} = +1.10 \text{ V}$; net: $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$.

Common Error

Multiplying half-reactions by coefficients and incorrectly multiplying E° values (E° is an intensive property).

Exam Strategy

Never scale E° values. Balance electrons by scaling the *equations*, but keep E° potentials unchanged.

Question 5**Student Task**

A galvanic cell uses $\text{Mg(s)}|\text{Mg}^{2+}(\text{aq})$ and $\text{Ag}^+(\text{aq})|\text{Ag(s)}$. Given $E^\circ(\text{Ag}^+/\text{Ag})=+0.80 \text{ V}$ and $E^\circ(\text{Mg}^{2+}/\text{Mg})=-2.37 \text{ V}$:

- (a) Determine anode and cathode.
- (b) Write cell notation.
- (c) State the direction of electron flow and ion migration (cation/anion) through the salt bridge.

Fully Worked Solution

Step 1 — Identification: Compare E°_{red} values: Ag^+/Ag (+0.80) higher than Mg^{2+}/Mg (−2.37).

Step 2 — Principle Statement: Higher E°_{red} is cathode (reduction). Anode is oxidation. Cell notation: anode | anode ion || cathode ion | cathode.

Step 3 — Explicit Reasoning: Cathode (reduction): $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$. Anode (oxidation): $\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$.
Cell notation: $\text{Mg(s)} | \text{Mg}^{2+}(\text{aq}) || \text{Ag}^+(\text{aq}) | \text{Ag(s)}$.

Electron flow: from anode to cathode externally $\rightarrow$ electrons flow from Mg electrode to Ag electrode.

Salt bridge migration: anions move toward anode compartment (to balance build-up of Mg^{2+}). Cations move toward cathode compartment (to replace Ag^+ being consumed / balance excess anions).

Step 4 — Validation: Oxidation at anode, reduction at cathode consistent; electron flow from Mg (strong reducing agent) to Ag^+ .

Step 5 — Final Answer: Anode = Mg, cathode = Ag; notation $\text{Mg} | \text{Mg}^{2+} || \text{Ag}^+ | \text{Ag}$; e^- flow $\text{Mg} \rightarrow \text{Ag}$; anions $\rightarrow$ anode, cations $\rightarrow$ cathode.

Common Error

Saying electrons travel through the salt bridge (they do not; ions do).

Exam Strategy

Memorize: *electrons* move in the wire; *ions* move in the salt bridge to maintain electrical neutrality.

Question 6

Student Task

Electroplating copper: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu(s)}$.

A current of 0.325 A runs for 3.00 h in a CuSO_4 solution.

Calculate the mass of Cu(s) deposited (g). Use $F = 9.65 \times 10^4 \text{ C} \cdot \text{mol}^{-1} \text{e}^-$ and $M(\text{Cu}) = 63.55 \text{ g} \cdot \text{mol}^{-1}$.

Fully Worked Solution

Step 1 — Identification: $I = 0.325 \text{ A} = 0.325 \text{ C} \cdot \text{s}^{-1}$; $t = 3.00 \text{ h} = 3.00 \times 3600 \text{ s} = 1.080 \times 10^4 \text{ s}$; $n(\text{e}^-) = Q/F$; Cu requires 2e^- per Cu.

Step 2 — Principle Statement: Charge $Q = It$. Moles of electrons $n(\text{e}^-) = Q/F$. Stoichiometry links electrons to moles Cu: $n(\text{Cu}) = n(\text{e}^-)/2$. Mass = $n(\text{Cu}) \times M$.

Step 3 — Explicit Reasoning: $Q = (0.325)(1.080 \times 10^4) = 3510 \text{ C}$.

$n(e^-) = 3510 / (9.65 \times 10^4) = 3.64 \times 10^{-2} \text{ mol } e^-$.

$n(\text{Cu}) = (3.64 \times 10^{-2}) / 2 = 1.82 \times 10^{-2} \text{ mol Cu}$.

$m(\text{Cu}) = (1.82 \times 10^{-2})(63.55) = 1.16 \text{ g}$.

Step 4 — Validation: Small current over a few hours gives gram-scale plating—reasonable. Units: $\text{C} \rightarrow \text{mol } e^- \rightarrow \text{mol Cu} \rightarrow \text{g}$.

Step 5 — Final Answer: $m(\text{Cu}) = 1.16 \text{ g}$ deposited.

Common Error

Forgetting to convert hours to seconds or forgetting the $2e^-$ per Cu^{2+} stoichiometry.

Exam Strategy

Write the chain explicitly: $I, t \rightarrow Q \rightarrow \text{mol } e^- \rightarrow \text{mol metal} \rightarrow \text{mass}$. Put the electron coefficient beside the half-reaction before you start.

Question 7

Student Task

Electrolysis produces chlorine at the anode: $2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2e^-$.

If a current of 2.50 A runs, how long (minutes) is required to produce 3.00 L of $\text{Cl}_2(\text{g})$ at STP?

Use molar volume $22.4 \text{ L} \cdot \text{mol}^{-1}$ and $F = 9.65 \times 10^4 \text{ C} \cdot \text{mol}^{-1} e^-$.

Fully Worked Solution

Step 1 — Identification: $V(\text{Cl}_2) = 3.00 \text{ L}$ at STP $\rightarrow n(\text{Cl}_2) = V/22.4$. Cl_2 requires 2 mol e^- per 1 mol Cl_2 . $Q = n(e^-)F$.
 $t = Q/I$.

Step 2 — Principle Statement: Convert gas volume $\rightarrow$ moles; use electron stoichiometry; then convert moles $e^- \rightarrow$ charge; then charge $\rightarrow$ time using $Q=It$.

Step 3 — Explicit Reasoning: $n(\text{Cl}_2) = 3.00/22.4 = 0.134 \text{ mol}$.

$n(e^-) = (0.134 \text{ mol } \text{Cl}_2)(2 \text{ mol } e^- / 1 \text{ mol } \text{Cl}_2) = 0.268 \text{ mol } e^-$.

$Q = (0.268)(9.65 \times 10^4) = 2.59 \times 10^4 \text{ C}$.

$$t = Q/I = (2.59 \times 10^4 \text{ C}) / (2.50 \text{ C} \cdot \text{s}^{-1}) = 1.04 \times 10^4 \text{ s}.$$

Convert to minutes: $(1.04 \times 10^4 \text{ s})(1 \text{ min}/60 \text{ s}) = 173 \text{ min}.$

Step 4 — Validation: Larger charge/time expected for gas production; minutes scale fits 2.5 A current.

Step 5 — Final Answer: $t \approx 173 \text{ min}.$

Common Error

Using $24.8 \text{ L} \cdot \text{mol}^{-1}$ (SATP) when the question states STP, or forgetting the $2e^-$ per Cl_2 .

Exam Strategy

Always circle the condition (STP vs SATP) and write the electron coefficient from the half-reaction beside your mole conversion.

Question 8

Student Task

Redox titration: In acidic solution, MnO_4^- oxidizes Fe^{2+} to Fe^{3+} .

A 25.00 mL sample of Fe^{2+} solution is titrated with $0.02000 \text{ mol} \cdot \text{L}^{-1} \text{ KMnO}_4$. The endpoint occurs at 18.60 mL.

Determine the concentration of Fe^{2+} ($\text{mol} \cdot \text{L}^{-1}$).

Fully Worked Solution

Step 1 — Identification: Need mole ratio from balanced equation in acid: $\text{MnO}_4^- + 8\text{H}^+ + 5\text{Fe}^{2+} \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O} + 5\text{Fe}^{3+}.$

Given: $C(\text{MnO}_4^-) = 0.02000 \text{ M}$, $V = 0.01860 \text{ L}$, $V(\text{Fe}^{2+}) = 0.02500 \text{ L}.$

Step 2 — Principle Statement: $n = CV$. Use stoichiometric ratio from balanced equation: 1 mol MnO_4^- reacts with 5 mol $\text{Fe}^{2+}.$

Step 3 — Explicit Reasoning: $n(\text{MnO}_4^-) = (0.02000)(0.01860) = 3.720 \times 10^{-4} \text{ mol}.$

$n(\text{Fe}^{2+}) = (3.720 \times 10^{-4})(5) = 1.860 \times 10^{-3} \text{ mol}.$

$C(\text{Fe}^{2+}) = n/V = (1.860 \times 10^{-3}) / (0.02500) = 0.07440 \text{ mol} \cdot \text{L}^{-1}.$

Step 4 — Validation: Concentration is plausible for titration volumes; sig figs consistent (4).

Step 5 — Final Answer: $[\text{Fe}^{2+}] = 0.07440 \text{ M}$.

Common Error

Using a 1:1 mole ratio instead of extracting 5:1 from the balanced redox equation.

Exam Strategy

Always write (or recall) the balanced redox equation first; titrations are stoichiometry problems, not 'plug-and-chug'.

Question 9**Student Task**

Electrolysis of aqueous NaCl (brine) with inert electrodes:

- (a) Predict the products at the cathode and anode using standard reduction potentials.
- (b) Explain why the predicted anode product can differ from the observed product in real brine electrolysis.

Fully Worked Solution

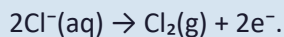
Step 1 — Identification: Aqueous brine contains Na^+ , Cl^- , and H_2O . Inert electrodes. Competing half-reactions include water reduction/oxidation and chloride oxidation.

Step 2 — Principle Statement: The most likely cathode reaction is the reduction with the more favorable (less negative / more positive) potential among available species; similarly for anode considering oxidation (reverse of listed reductions). Kinetics/overpotential and concentration can change which reaction dominates.

Step 3 — Explicit Reasoning: Cathode (reduction): $\text{Na}^+ + \text{e}^- \rightarrow \text{Na(s)}$ is highly unfavorable in water; instead water is reduced:



Anode (oxidation): Candidates are oxidation of Cl^- to Cl_2 and oxidation of water to O_2 . In concentrated brine, chloride oxidation often dominates:



- (a) Predicted products: cathode $\text{H}_2\text{(g)} + \text{OH}^-\text{(aq)}$; anode $\text{Cl}_2\text{(g)}$ (often).

(b) Why prediction can differ: Even if water oxidation is thermodynamically possible, overpotential for O_2 evolution, electrode material effects, and high $[Cl^-]$ can favor Cl_2 formation; in dilute solutions or different electrodes, O_2 may form instead. Thus predicted (thermo) vs actual (kinetics/conditions) can differ.

Step 4 — Validation: Observed industrial chlor-alkali process produces H_2 at cathode and Cl_2 at anode; matches reasoning under concentrated brine conditions.

Step 5 — Final Answer: Cathode: $H_2 + OH^-$ from water reduction; Anode: typically Cl_2 from chloride oxidation; deviations occur due to overpotential, concentration, and electrode/kinetic effects.

Common Error

Claiming 'Na metal forms at the cathode' in aqueous solution (ignores competition with water).

Exam Strategy

Before choosing a half-reaction, list *all species present* (including water) and consider which reduction is realistically favored under aqueous conditions.

Question 10

Student Task

Structured compare-and-contrast (extended response): Compare a voltaic cell and an electrolytic cell.

Your answer must address: (i) spontaneity and energy conversion, (ii) anode/cathode processes and electrode signs, (iii) electron/ion movement, (iv) one real-world application of each, and (v) one safety/environment consideration (STS).

Fully Worked Solution

Step 1 — Identification: Two cell types: voltaic (galvanic) vs electrolytic; both involve redox with oxidation at anode and reduction at cathode.

Step 2 — Principle Statement: Voltaic cells convert chemical potential energy to electrical energy via a spontaneous redox reaction ($E_{cell} > 0$). Electrolytic cells use electrical energy from an external power supply to drive a nonspontaneous redox reaction ($E_{cell} < 0$ without the power supply).

Step 3 — Explicit Reasoning:

(i) Spontaneity/energy: Voltaic is spontaneous (produces voltage); electrolytic is nonspontaneous (consumes electrical energy).

(ii) Anode/cathode: In BOTH: oxidation at anode, reduction at cathode. Signs differ: voltaic anode is negative (source of e^-), cathode positive; electrolytic anode positive (pulled by power supply), cathode negative.

(iii) Movement: Electrons move through the external circuit from anode to cathode in both; ions migrate through electrolyte/salt bridge to maintain neutrality (anions toward anode, cations toward cathode).

(iv) Applications: Voltaic—batteries/fuel cells (portable power). Electrolytic—electroplating, refining/electrowinning metals, chlor-alkali process.

(v) STS: Voltaic—battery disposal (heavy metals, recycling logistics). Electrolytic—chlorine production and handling hazards; energy use and emissions depend on electricity source. Corrosion control (sacrificial anodes) prevents infrastructure failure and environmental releases.

Step 4 — Validation: Core invariants (oxidation at anode, reduction at cathode) maintained; differences correctly tied to spontaneity and electrode sign conventions.

Step 5 — Final Answer: Voltaic = spontaneous chemical $\rightarrow$ electrical with negative anode/positive cathode; Electrolytic = driven electrical $\rightarrow$ chemical with positive anode/negative cathode; both follow oxidation at anode, reduction at cathode; include applications and STS considerations as above.

Common Error

Stating 'anode is negative' as a universal rule (true only for voltaic, not electrolytic).

Exam Strategy

Anchor your response to invariants first (oxidation at anode, reduction at cathode), then layer on the differences (spontaneous vs driven, electrode signs, applications, STS).